

Algorithms

Lecture #38

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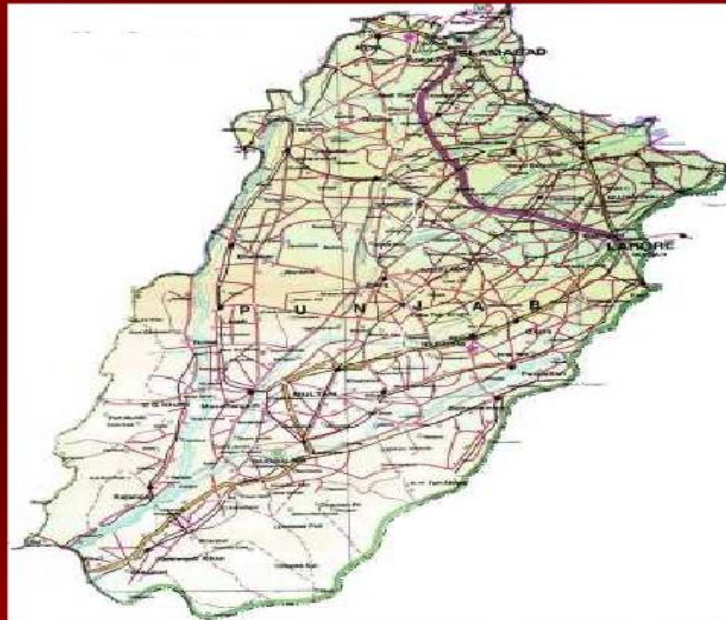
Shortest Path

Shortest Path

- A motorist wishes to find the shortest possible route between Peshawar and Karachi.
- Given a road map of Pakistan on which the distance between each pair of adjacent cities is marked
- Can the motorist determine the shortest route?

Shortest Path

Shortest Paths



Shortest Path

Shortest Paths

In the shortest-paths problem

- We are given a weighted, directed graph $G = (V, E)$
- The weight of path $p = \langle v_0, v_1, \dots, v_k \rangle$ is the sum of the constituent edges:

$$w(p) = \sum_{i=1}^k w(v_{i-1}, v_i)$$

Shortest Path

Shortest Paths

We define the ***shortest-path weight*** from u to v by

$$\delta(u, v) = \begin{cases} \min\{w(p) : u \overset{p}{\rightsquigarrow} v\} & \text{if there is a path from } u \text{ to } v \\ \infty & \text{otherwise} \end{cases}$$

Shortest Path

Shortest Path

- Problems such as shortest route between cities can be solved efficiently by modelling the road map as a graph.
- The nodes or vertices represent cities and edges represent road.
- Edge weights can be interpreted as distances. Other metrics can also be used, e.g., time, cost, penalties and loss.

Shortest Path

Shortest Path

- Similar scenarios occur in computer networks like the Internet where data packets have to be routed.
- The vertices are routers.
- Edges are communication links which may be be wire or wireless.
- Edge weights can be distance, link speed, link capacity link delays, and link utilization.

Shortest Path

Shortest Path

- The breadth-first-search algorithm we discussed earlier is a shortest-path algorithm that works on un-weighted graphs.
- An un-weighted graph can be considered as a graph in which every edge has weight one unit.

Shortest Path

Shortest Path Variants

Single-source shortest-path problem:

- Find shortest paths from a given (single) source vertex $s \in V$ to every other vertex $v \in V$ in the graph G .

Shortest Path

Shortest Path

Single-destination shortest-paths problem:

- Find a shortest path to a given destination vertex t from each vertex v .
- We can reduce this problem to a single-source problem by reversing the direction of each edge in the graph.

Shortest Path

Shortest Path

Single-pair shortest-path problem:

- Find a shortest path from u to v for given vertices u and v .
- If we solve the single-source problem with source vertex u , we solve this problem also.
- No algorithms for this problem are known to run asymptotically faster than the best single-source algorithms in the worst case.

Shortest Path

Shortest Path

All-pairs shortest-paths problem:

- Find a shortest path from u to v for every pair of vertices u and v .
- Although this problem can be solved by running a single-source algorithm once from each vertex, it can usually be solved faster.

Negative Edge Weights

Negative Edge Weights

Dijkstra's algorithm

- Dijkstra's algorithm is a simple greedy algorithm for computing the single-source shortest-paths to all other vertices.
- Dijkstra's algorithm works on a weighted directed graph $G = (V, E)$ in which all edge weights are non-negative,
- i.e., $w(u, v) \geq 0$ for each edge $(u, v) \in E$.

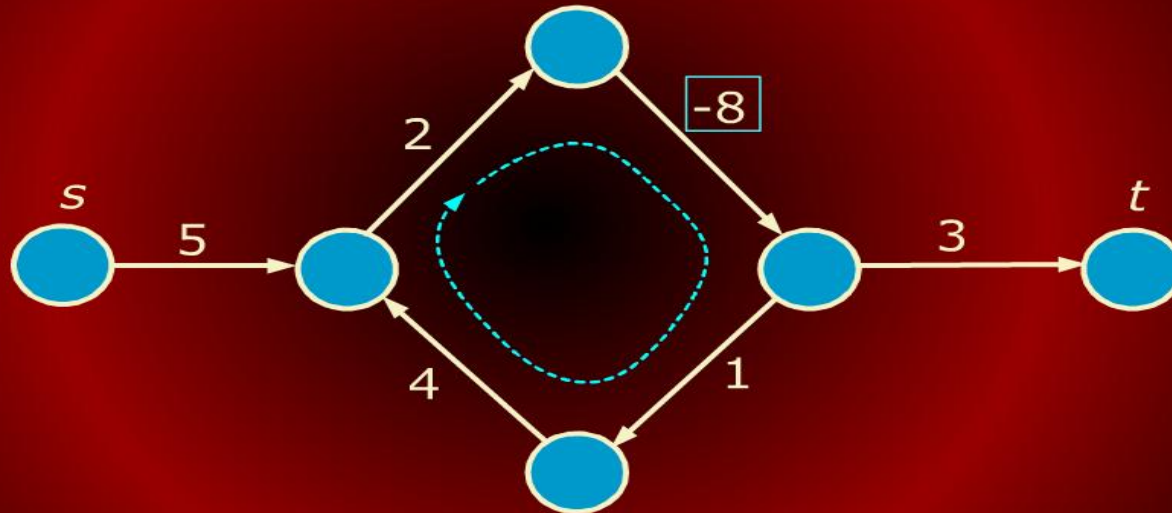
Negative Edge Weights

Negative Edge Weights

- Negative edges weights maybe counter to intuition but this can occur in real life problems.
- However, we will ***not allow negative cycles*** because then there is no shortest path.
- If there is a negative cycle between, say, s and t , then we can always find a shorter path by going around the cycle one more time.

Negative Edge Weights

Negative Edge Weights



Dijkstra's Algorithms

Dijkstra's Algorithm

- The basic structure of Dijkstra's algorithm is to maintain an estimate of the shortest path from the source vertex to each vertex in the graph.
- Call this estimate $d[v]$.
- Intuitively, $d[v]$ will be the length of the shortest path that the *algorithm knows of* from s to v .

Dijkstra's Algorithms

Dijkstra's Algorithm

- This value will always be greater than or equal to the true shortest path distance from s to v .
- I.e., $d[v] \geq \delta(u, v)$.
- Initially, we know of no paths, so $d[v] = \infty$
- Moreover, $d[s] = 0$ for the source vertex.

Dijkstra's Algorithms

Dijkstra's Algorithm

- As the algorithm goes on and sees more and more vertices, it attempts to update $d[v]$ for each vertex in the graph.
- The process of updating estimates is called *relaxation*.
- The algorithm stops when all $d[v]$ values converge to the true shortest distances.

Shortest Path